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THE NATIONAL COLLEGE BASAVANAGUDI, BENGALURU- 560 004

AUTONOMOUS NAAC Accredited 'A' Grade

Affiliated to the Bengaluru City University Website: www.ncbgudi.com

B.Sc. Course Matrix Structure for SEP 2024

Semester	Discipline components	Papers	Credits	Hours /week	MARKS
I	DSC1-EL-T1	ANALOG AND DIGITAL ELECTRONICS-I	3	4	100
	CORE-2		3	4	100
	CORE-3		3	4	100
	DSC1-EL-P1	ANALOG AND DIGITAL ELECTRONICS-I Lab	2	3	50
	LAB-2		2	3	50
	LANG-1	LANGUAGE –KAN/SAN/HIN	3	4	100
	LANG-2	GENERIC ENGLISH	3	4	100
	Compulsory paper	Constitutional Values	3	4	100
		Total credits	22		700

Semester	Discipline components	Papers	Credits	Hours /week	MARKS
II	DSC1-EL-T2	ANALOG AND DIGITAL ELECTRONICS-II	3	4	100
	CORE-2		3	4	100
	CORE-3		3	4	100
	DSC1-EL-P2	Device Modelling & Characterization Lab (TCAD Based)	2	3	50
	LAB-2	ANALOG AND DIGITAL ELECTRONICS-II Lab	2	3	50
	LANG-1	LANGUAGE –KAN/SAN/HIN	3	4	100
	LANG-2	GENERIC ENGLISH	3	4	100
	Compulsory paper	EVS	2	3	50
		Total credits	21		650

Semester	Discipline components	Papers	Credits	Hours/week	MARKS
III	CORE-1	EMBEDDED C AND DIGITAL DESIGN USING VERILOG	3	4	100
	CORE-2		3	4	100
	CORE-3		3	4	100
	DSE	POWER ELECTRONICS AND ELECTRIC VEHICLES	2	2	50
	LAB-1	EMBEDDED C AND DIGITAL DESIGN USING VERILOG Lab	2	3	50
	LAB-2		2	3	50
	LANG-1	LANGUAGE –KAN/SAN/HIN	3	4	100
	LANG-2	GENERIC ENGLISH	3	4	100
		Total credits	21		650

Semester	Discipline components	Papers	Credits	Hours/week	MARKS
IV	CORE-1	ADVANCED COMMUNICATION SYSTEMS	3	4	100
	CORE-2		3	4	100
	CORE-3		3	4	100
	DSE	MEDITRONICS	2	2	50
	LAB-1	ADVANCED COMMUNICATION SYSTEMS Lab	2	3	50
	LAB-2		2	3	50
	SEC	Skill Lab: Project	2	3	50
	LANG-1	LANGUAGE –KAN/SAN/HIN	3	4	100
	LANG-2	GENERIC ENGLISH	3	4	100
		Total credits	23		700

B.Sc. Course Matrix Structure for NEP 2020

Semester	Discipline components	Papers	Credits	Hours /week	MARKS
V	BS5-EL-CT5	Communication II	4	4	100
	BS5-EL-CT6	Embedded Controllers	4	4	100
	BS5-XX-CT5		4	4	100
	BS5-XX-CT6		4	4	100
	BS5-EL-CP5	Communication II Lab	2	4	50
	BS5-EL-CP6	Embedded Controllers Lab	2	4	50
	BS5-XX-CP5		2	4	50
	BS5-XX-CP6		2	4	50
	SEC	Employability skill	2	3	50
		Total credits	26		650

Semester	Discipline components	Papers	Credits	Hours /week	MARKS
VI	BS5-EL-CT7	Electronic Instrumentation and Biomedical Instruments	4	4	100
	BS5-EL-CT8	Instrumentation and IoT Lab	4	4	100
	BS5-XX-CT7		4	4	100
	BS5-XX-CT8		4	4	50
	BS5-EL-CP7	Instrumentation and IoT Lab	2	4	50
	BS5-EL-CP8	Project Lab	2	4	50
	BS5-XX-CP7		2	4	50
	BS5-XX-CP8		2	4	50
		Internship	2		50
		Total credits	24		600